

Linear First-Order Equations & the Integrating Factor



Recognizing linear form

- 1 Write $y' + 2xy = x$ in the standard form $y' + P(x)y = Q(x)$ and identify $P(x)$ and $Q(x)$.
 - 2 Is $y' = \frac{y}{x} + x^2$ linear? Rewrite it in standard form.
 - 3 Is $y' = y^2 + x$ a linear first-order equation? Justify your answer.
 - 4 Write $xy' + 3y = \sin x$ ($x > 0$) in standard form.
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Finding the integrating factor

- 5 Find the integrating factor for $y' + 3y = e^x$.
 - 6 Find the integrating factor for $y' - \frac{2}{x}y = x$, $x > 0$.
 - 7 Find the integrating factor for $y' + \frac{1}{x}y = 3x$, $x > 0$.
 - 8 Find the integrating factor for $y' + 2xy = 4x$.
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Solving with the integrating factor

- 9 Solve $y' + 3y = e^x$.
 - 10 Solve $y' - \frac{2}{x}y = x$, $x > 0$.
 - 11 Solve $y' + \frac{1}{x}y = 3x$, $x > 0$.
 - 12 Solve $y' + 2xy = 4x$.
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Initial value problems

- 13 Solve $y' + y = 2$, $y(0) = 0$.
- 14 Solve $y' + 2y = 4x$, $y(0) = 1$.
- 15 Solve $y' - y = e^{2x}$, $y(0) = 3$.

16 Solve $y' + \frac{2}{x}y = \frac{\cos x}{x^2}$, $x > 0$, $y(\pi) = 0$.

Application: a filling tank

- 17 A tank starts with 100 L of pure water. Brine with concentration 2 kg/L flows in at 3 L/min, and the well-mixed solution flows out at the same rate. If $Q(t)$ is the salt (kg) at time t , the balance gives $Q' = 6 - 0.03Q$. Solve for $Q(t)$ given $Q(0) = 0$.
- 18 For the tank above, what is the long-run (as $t \rightarrow \infty$) salt content, and why does that match the tank's incoming brine concentration times its volume?
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Application: an RL electrical circuit

- 19 An RL circuit with resistance R , inductance L , and constant EMF E_0 satisfies $L \frac{dI}{dt} + RI = E_0$. Rewrite this in standard linear form and identify its integrating factor.
- 20 Solve the RL circuit equation from above for $I(t)$ given $I(0) = 0$.
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Mixed review

- 21 Explain why every first-order linear equation $y' + P(x)y = Q(x)$ can, in principle, be solved by the integrating-factor method, regardless of the specific P and Q .
- 22 Solve $y' - 4y = 8x$.
- 23 Solve $\cos x y' + \sin x y = 1$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$. (Divide by $\cos x$ first, and recognize the integrating factor without computing an integral.)
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Comparing linear equations with separable equations

- 24 The equation $y' = xy$ is both separable and linear (with $Q(x) = 0$ — a homogeneous linear equation). Solve it using the integrating-factor method and confirm it matches the separable-equation approach.
- 25 For which choice(s) of $Q(x)$ does a linear equation $y' + P(x)y = Q(x)$ reduce to a separable equation in general? Explain.
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Working backward from a solution

26 A linear equation $y' + P(x)y = Q(x)$ has general solution $y = x^2 + \frac{C}{x}$. Given that $P(x) = \frac{1}{x}$, find $Q(x)$.